Drones, Deadlines and NP-Hardness

Metaheuristics for a Capacitated Medical Delivery Network

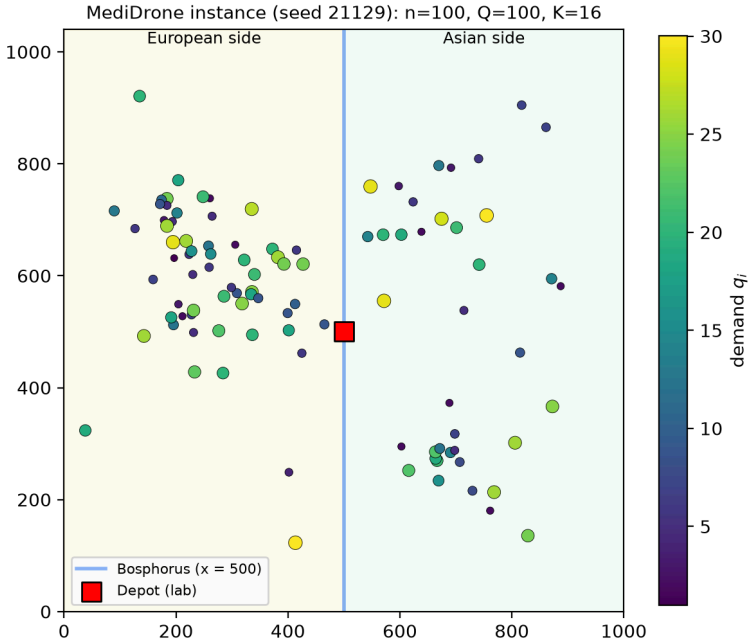
[Member 1 · Member 2 · Member 3 · Member 4]

CMP 3005 – Analysis of Algorithms · Fall 2026–27 · Dr. Günet Eroğlu

The problem: MediDrone Istanbul

- A central lab (depot) and clinics on both sides of the Bosphorus
- Each drone carries at most Q; every clinic is served **exactly once**; at most K drones
- Minimise the total flight distance
- Crossing the strait costs $+P = 150$
- This is the **Capacitated Vehicle Routing Problem (CVRP)**

Our team instance: seed $6936 + 2546 + 6698 + 4949 = 21129$ · $n = 100$, $Q = 100$, $K = 16$



Why is it hard? Two NP-hard problems in one

- **CVRP-DEC $\in$ NP**: certificate = the routes, verified in $O(n)$
- **Reduction chain**: Hamiltonian Cycle $\underset{p}{\leq}$ metric TSP (weights 1/2 keep the triangle inequality) $\underset{p}{\leq}$ CVRP ($K = 1$, $q_i = 1$, $Q = n$)
- Composing polynomial reductions is polynomial $\Rightarrow$ CVRP is NP-complete
- **Hidden problem**: "do the customers fit into K vehicles?" = **Bin Packing** $\Rightarrow$ strongly NP-complete
  - even **finding a feasible solution** is hard: our benchmarks are 92–99 % full
  - Clarke–Wright returns 7 routes on A-n45-k6 ($K = 6$)!
- **Bosphorus penalty keeps the metric**: a crossing path $i \to j \to k$ crosses at least once (it is a sum of two metrics)

Three algorithms, all our own code

Clarke–Wright

- savings $s_{ij} = d_{0i} + d_{0j} - d_{ij}$
- greedy merges in decreasing order
- $O(n^2 \log n)$, deterministic
- start solution for GA and TS

Memetic GA

- giant tour + **Split** DP (shortest path in a DAG, $O(nL)$)
- fleet-limited, soft-capacity Split
- OX, inversion, tournament, $(\mu + \lambda)$, no clones
- granular local search on 50 % of the children

Tabu Search

- relocate, swap, 2-opt, 2-opt*
- $O(1)$ delta evaluation
- 20 nearest-neighbour candidate list
- arc tabu list, aspiration
- adaptive penalty $\lambda \cdot$ overload

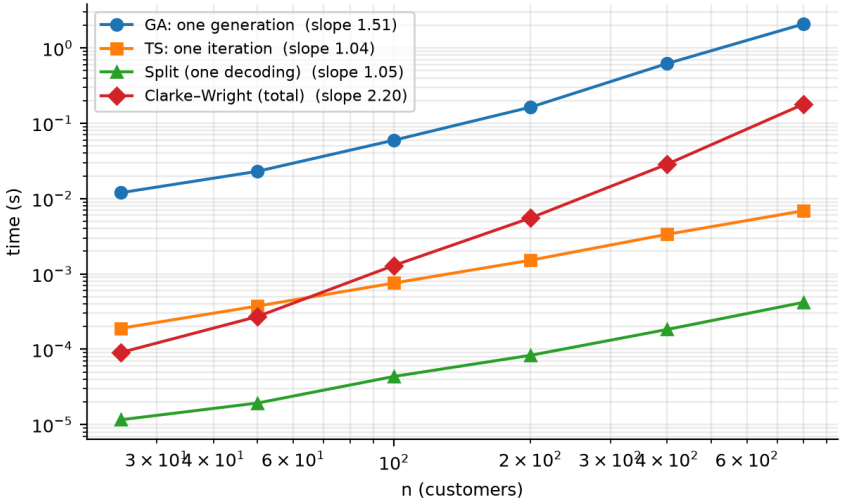
Split: decoding a giant tour optimally

- Node i = "first i customers served"
- Arc (i, j) = one trip $T_{i+1}...T_j$ if it fits in Q
- Shortest $0 \to n$ path = optimal cut
- $V(j) = \min_i V(i) + c(i, j)$
- Stop extending j once load $> Q$: $O(nL)$ instead of $O(n^2)$
- Layered version for $\leq K$ routes: $O(KnL)$

```
for i = 0..n-1:
  load = 0
  for j = i+1..n:
    load += q[T_j]
    if load > Q: break
    cost = update in O(1)
    V[j] = min(V[j], V[i] + cost)
```

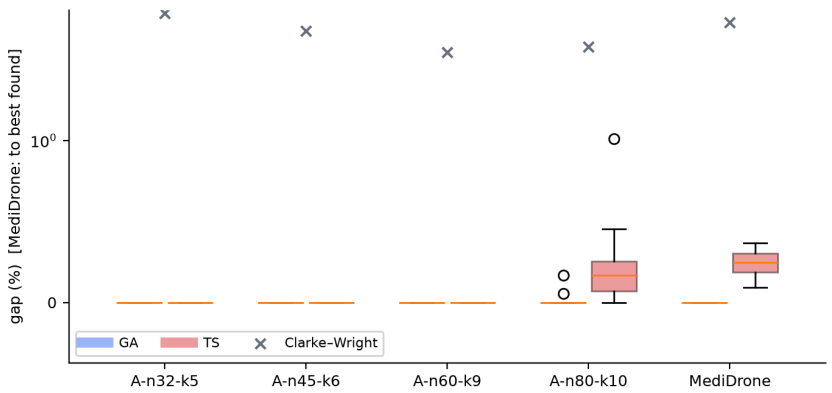
Complexity: theory vs. measurement

- TS iteration: $\Theta(nk)$ scan + $O(n)$ apply $\Rightarrow$ $\Theta(n)$
  - measured slope **1.04**
- GA generation: Split $\Theta(KnL)$ + local search $\Theta(ng)$
  - measured slope **1.51**
- Delta evaluation: $O(1)$ vs. $\Theta(n)$
  - $124\times$ faster at $n = 800$



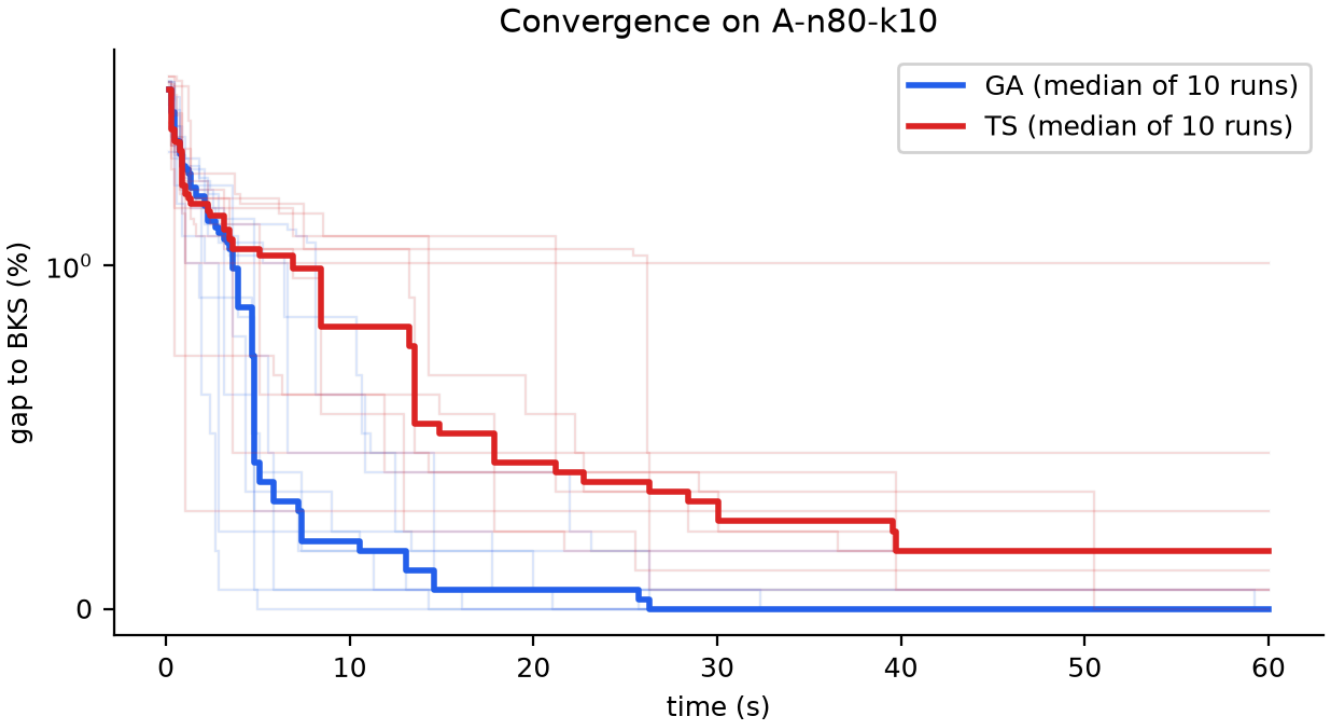
Results (10 seeds, equal time budget)

Instance	BKS	CW gap %	GA best / mean gap %	TS best / mean gap %	GA std	TS std
A-n32-k5	784	7.4	0 / 0	0 / 0	0	0
A-n45-k6	944	5.61	0 / 0	0 / 0	0	0
A-n60-k9	1354	3.99	0 / 0	0 / 0	0	0
A-n80-k10	1763	4.37	0 / 0.02	0 / 0.25	0.97	5.3

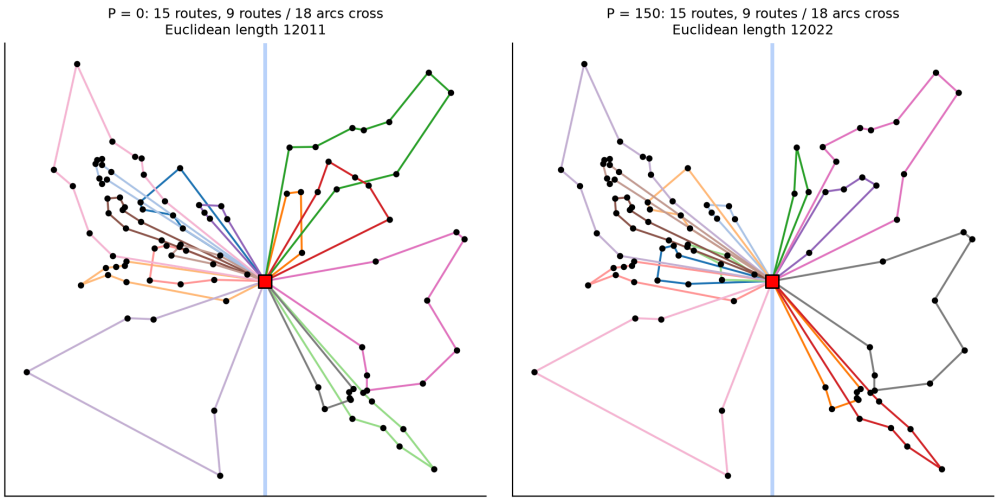


- Tiny instances: both methods reach the DP optimum in all runs
- Every reported solution passes the independent checker
- Full table, convergence and statistics are in the report

Convergence

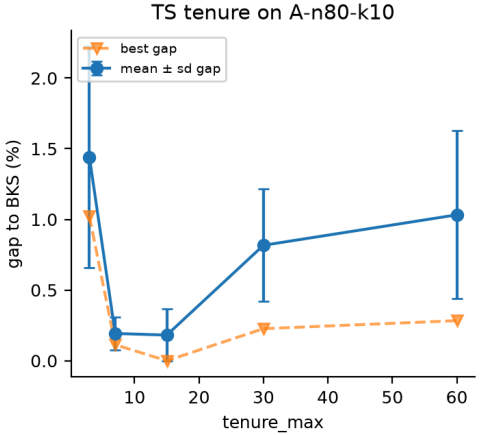


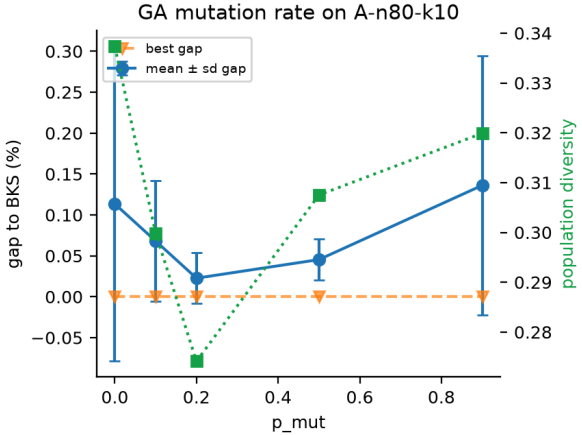
MediDrone plans and the Bosphorus

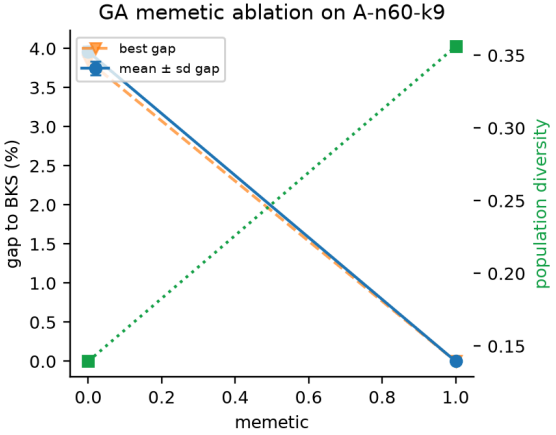


- With $P = 0$: 9 routes cross the strait (18 crossing arcs)
- With $P = 150$: 9 routes cross (18 arcs)
- Same structure in both: 9 Europe-only + 6 Asia-only routes, no mixed route
- Depot on the strait $\Rightarrow$ every route that visits Europe crosses exactly twice
- Penalty $= 300 \times \#$ European routes $\geq 300 \cdot \lceil 875/100 \rceil = 2700$, and our plans attain this bound
- Euclidean length: 12011 vs. 12022

Parameter sensitivity







Takeaways

- CVRP is NP-hard twice over (TSP **and** Bin Packing), yet good plans take seconds
- Delta evaluation and candidate lists are what make Python fast enough
- Allowing infeasibility with an adaptive penalty is essential on nearly full instances
- Memetic GA (Split + local search) is the most robust; TS is very fast early on
- No optimality guarantee, but ≤ 1 % from the best-known solutions, checked independently
- Next steps: hybrid GA + TS, adaptive large neighbourhood search, time windows for vaccine deliveries

Thank you — questions?